B2.1 Membranes and membrane transport

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-b2-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 237–243 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Membrane fluidity

B2.1.11 Fatty acids and fluidity

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Saturated fatty acid tails are straight and pack closely, so they make a membrane less fluid. Unsaturated tails are kinked at their double bonds, cannot pack closely and keep the membrane fluid at lower temperatures. Membranes must be fluid enough for proteins to move and work, but not so fluid that they leak. Organisms such as the lake sturgeon raise the share of unsaturated fatty acids when the water gets colder: **homeoviscous adaptation**.

B2.1.12 Cholesterol

pp. 237–238

In animal membranes, cholesterol sits with its polar **-OH group** next to the phosphate heads and its non-polar rings and tail among the fatty acid tails. It acts as a **fluidity buffer**: at low temperature it keeps phospholipids apart so the membrane stays fluid; at high temperature it restricts their movement so the membrane is less fluid and less leaky.

B2.1.13 Vesicles form and fuse

pp. 238–239

Because phospholipids can flow, a membrane can pinch inwards to make a vesicle (**endocytosis**: phagocytosis of solids, pinocytosis of liquids) and a vesicle can fuse with the plasma membrane (**exocytosis**) without leaving a hole. Both need ATP. Macrophages engulf old red blood cells; plasma cells secrete antibodies and neurons release neurotransmitters by exocytosis.

Channels, pumps and cotransporters

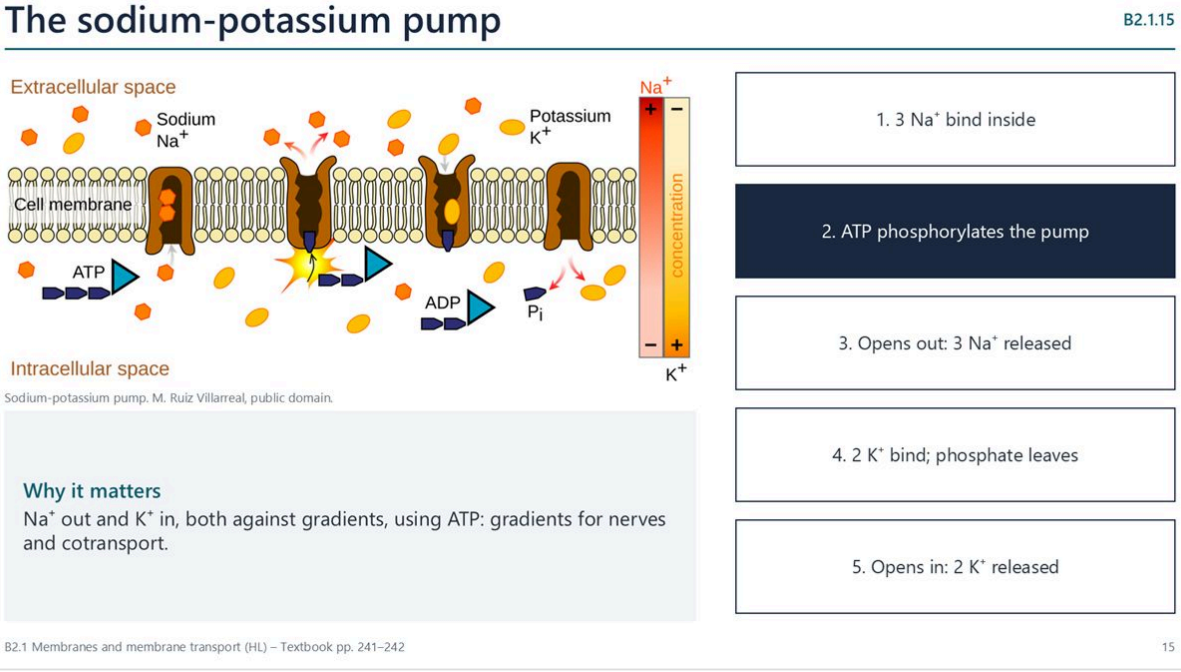
B2.1.14 Gated ion channels

pp. 239–241

Gated channels open and close, letting ions diffuse down their gradients. **Voltage-gated** Na^+ and K^+ channels in axons open when the membrane potential changes; a K^+ channel is then plugged by a “ball and chain” until the membrane repolarizes. **Neurotransmitter-gated** channels, such as the nicotinic acetylcholine receptor, open when acetylcholine binds, letting Na^+ into the post-synaptic cell.

B2.1.15 The sodium-potassium pump

pp. 241–242



Sodium-potassium pump. M. Ruiz Villarreal, public domain.

An **exchange transporter**. Three Na^+ bind inside; ATP phosphorylates the pump, which opens outwards and releases them; two K^+ bind outside; the phosphate leaves and the pump opens inwards, releasing K^+. Both ions move against their gradients, building the gradients used by neurons and cotransporters.

B2.1.16 Indirect active transport of glucose

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In the small intestine and kidney tubules, Na^+/K^+ pumps keep Na^+ low inside the cell. Na^+ diffuses back in through a **sodium-dependent glucose cotransporter**, carrying glucose with it (two Na^+ per glucose), even against the glucose gradient. The cotransporter uses no ATP itself, so this is **indirect active transport**. Glucose then leaves the cell by facilitated diffusion into the blood.

Tissues

B2.1.17 Cell adhesion

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Cell-adhesion molecules (CAMs) are membrane proteins that bind cells to each other (usually **cadherins**) or to the extracellular matrix (usually **integrins**). They form junctions: **tight junctions** seal epithelia, **anchoring junctions** give strength and **gap junctions** link the cytoplasm of neighbouring cells. Plant cells rely instead on interlocking walls and plasmodesmata.

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